



Influence of executive functions on quality of life in Pediatric Epilepsy: A cross-sectional study

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ARTICLE INFO

KEYWORDS:

Executive functions
Quality of life
Epilepsy

ABSTRACT

Background: The EpiTrack Junior is a screening tool assessing executive function in children with epilepsy. This study aimed to investigate whether children and adolescents with epilepsy are at a higher risk of experiencing a reduced quality of life if they also reveal abnormal results reflecting executive dysfunction.

Methods: We screened patients for executive dysfunction using the clinical test tool EpiTrack Junior. To assess health-related quality of life (HRQoL), the German children's and parents' version of KINDL questionnaire was used. The KINDL scores (total score and dimensions scores) of patients with and without clinically conspicuous values were compared (≤ 28 and > 29 , respectively). In addition, the exact EpiTrack Junior point scores were correlated with the KINDL total score and the scores of all KINDL dimensions.

Results: In this study 112 (mean age = 11.72, SD = 3.6) patients with epilepsy and their parents were included. Patients with executive dysfunctions (EpiTrack Junior values ≤ 28) scored significantly poorer in the QoL categories 'family' and 'social environment' than patients without. In the 'family' dimension, the child-report revealed the following data: $z = -2.759$; adjusted p -value: 0.042, and in the 'friends' dimension, parent-reports yielded the following data: $z = -3.645$; adjusted p -value: 0.007. In contrast, the 'self-esteem' dimension in the children's version showed significantly higher values in patients with executive dysfunctions than for those without: $z = -2.524$; adjusted p -value: 0.042. No significant differences between patients with and without executive dysfunctions were found for the overall quality of life (as assessed by the KINDL 'total score') as well as for the other dimensions (school, physical and emotional well-being).

Conclusions: No differences were found in the overall quality of life between patients with and without executive dysfunction. Nevertheless, executive dysfunction appeared to have a negative impact on some areas of life, such as family and friends, and was a predictor of increased self-esteem.

Abbreviations: AWE, Adolescents with epilepsy; ASM, Anti-seizure medication; CI, Confidence interval; CWE, Children with epilepsy; EF, Executive functions; HRQoL, Health-related quality of life; Mdn, Median; SD, Standard deviation; ES, Effect size.

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<https://doi.org/10.1016/j.epilepsyres.2025.107646>

Received 10 April 2025; Received in revised form 18 June 2025; Accepted 29 August 2025

Available online 5 September 2025

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1. Introduction

Epilepsy is one of the most common, chronic neurological disorders in children (Ünalp et al., 2022). According to current guidelines, epilepsy is defined as an enduring predisposition to generate epileptic seizures (Reilly et al., 2011; Davies et al., 2003) and classified into subtypes based on the type of seizure, type of epilepsy, epilepsy syndromes, and etiology (Dunn et al., 2009). A large part of those epilepsy subtypes and syndromes are characterized by a childhood onset. Assuming a mean prevalence of 0.5 % of epilepsy in children and adolescents in the European Union, it can be estimated that at least 63.000 people with epilepsy are aged under 18 years in Germany (Riechmann et al., 2019; Forsgren et al., 2005; Noble et al., 2017).

Epilepsy is a disease affecting not only physical functions but is also often accompanied by comorbidities such as cognitive impairment, including executive dysfunction, intellectual and learning disability, as well as psychiatric and behavioral disorders (Dunn et al., 2009; Reuner et al., 2016; Fine and Wirrell, 2020; Schraegle and Titus, 2016). Disease-specific restrictions on leisure activities, mobility, career choice, and other social aspects, as well as frequent epilepsy-related accidents and the named accompanying comorbidities can affect one's perceived quality of life (Riechmann et al., 2019). Several studies have repeatedly shown that children and adolescents with epilepsy are at high risk of a reduced health-related quality of life compared to healthy populations and even compared to children with other chronic disorders such as asthma and diabetes (Ünalp et al., 2022; Schraegle and Titus, 2016; Miller et al., 2003; Austin et al., 2000). There is strong evidence for epilepsy-related factors, such as high medication load and high number of anti-seizure medications, having a negative impact on HRQoL. Behavioral disorders or family factors have a significant impact, too (Miller et al., 2003; Devinsky et al., 1999; Sherman et al., 2006). However, there is less study data on the role of executive functions in quality of life. The executive system is associated with higher-level skills such as cognitive flexibility, working memory, the ability to plan and organize situations, as well as behavioral and emotional self-regulation (Diamond, 2013; Seniów, 2012; Powell and Voeller, 2004). A dysfunction in this domain can cause increasing difficulties for the growing child in everyday life (Powell and Voeller, 2004). Considering the great importance of executive functions and the frequency with which impairments in this area occur in children with epilepsy (MacAllister et al., 2014), it becomes clear how important it is to address this topic in the context of childhood epilepsy.

Therefore, this explorative study aimed to identify a possible association between executive dysfunction and quality of life scores in children and adolescents with epilepsy and to find out which areas of life are particularly affected in patients with executive dysfunction.

In a cross-sectional study, we compared the QoL scores, measured using the KINDL questionnaire, of children with epilepsy with and without executive dysfunction using the EpiTrack^R Junior. We decided in favor of the EpiTrack as many previous studies on executive functions used questionnaires, while the EpiTrack required patients to perform tasks, which increases objectivity.

2. Materials and methods

2.1. Study design

This clinical cross-sectional non-randomized observational study was conducted at the Comprehensive Epilepsy Center for Children and Adolescents of Ludwig-Maximilians-University (LMU) Hospital. A total of 112 outpatient and inpatient children with epilepsy and their parents were prospectively recruited successively. Patients and parents were informed about the study and asked about consent. Further demographic data were also documented using information in the institutional database. Thus, all primary outcome parameters (EpiTrack Junior Scores and KINDL scores) were collected prospectively. Without

prior randomization or formation of a control group, all patients were combined into a single group. Patient's names were pseudonymized by a random code generated by MD5Hex.

Our study was approved by the local ethics committee of the LMU of Munich (reference: # 21 – 0887).

The study was conducted in accordance with the ethical principles of Helsinki and written informed consent was obtained from parents and patients.

2.2. Data collection

Patients had to fulfil the following criteria to be included in our study: confirmed epilepsy diagnosis, age between six and 17 years, basic knowledge of German or English and a current result for each the KINDL questionnaire and the EpiTrack Junior and a written informed consent by the parents/caregivers. Data on HRQoL and EF were not part of the routine clinical evaluation. Patients were prospectively enrolled for their eligibility for the study and asked for their consent to participate in the evaluation. We collected the questionnaires (KINDL) and tests (EpiTrack) from October 2021 to May 2024 and, where not already available, also carried out a CBCL questionnaire.

The Child Behavior Checklist (CBCL) is a standardized instrument used by parents to assess a wide range of emotional and behavioral problems in children and adolescents (Achenbach and Ruffle, 2000).

Data that was collected in addition to the patient's demographic data were: age at epilepsy onset, epilepsy duration, seizure frequency, etiology of epilepsy, seizure semiology, epilepsy syndrome, epilepsy surgery, additional neurologic or psychiatric diagnoses, current and previous ASMs, and already existing results of neuropsychological tests. According to the current guidelines of the International League Against Epilepsy, the seizure and epilepsy syndrome classifications were adapted (Specchio et al., 2022).

Exclusion criteria were: an unconfirmed epilepsy diagnosis, lack of German or English language skills, age < six and > 17 years, and severe mental impairment (in the sense of an intellectual disability, to an extent that makes it impossible to complete questionnaires and tests).

2.3. Executive functioning evaluation

For screening of executive dysfunction, the neuropsychological test 'EpiTrack Junior' was performed. The test was developed as a clinical screening tool for the rapid assessment and continuous monitoring of executive and attentional functions in children and adolescents with epilepsy (Helmstaedter et al., 2010). The EpiTrack Junior consists of six subtests that measure attention, reaction inhibition, visuomotor speed, planning, verbal fluency, and working memory (Fisher et al., 2005). To be able to take the test, one must be at least six years old, know the numbers from one to 20 and have a basic understanding of the German or English alphabet. You can receive a maximum of 7 points and a minimum of 1 point for each subtest. The test score is calculated using the following age correction: 6 years: + 14 points; 7 years: + 8 points; 8 years: + 4 points; 9 years: + 2 points; 10 years: no correction; 11 – 14 years: – 2 points; 15 – 18 years: – 4 points. Thus, the total test result can be between 6 and 56 points. The interpretation of the current test performance according to the publisher is: ≤ 28 P: significant impairment; 29–30 P: mild impairment; 31–35 P: on average; ≥ 36 P: well performed (Helmstaedter et al., 2010).

2.4. Evaluation of health-related quality of life

The HRQoL of children and adolescents with epilepsy was measured using the general scales of KINDL questionnaire, one child version, and one parent version. The general KINDL questionnaire can be used to assess HRQoL of any child aged older than three and therefore does not contain any epilepsy or disease-specific questions. Additional scales for selected diseases were not added. The decision was made since the

epilepsy-specific version is only available as a KidKINDL version and therefore only for ages 7–13 y. There is no parent version here either.

There are different versions depending on age: Kiddy: children 4–6 y; KidKINDL: children 7–13 y; Kiddo: adolescents 14–17 y; Kiddy-KINDL: parents 3–6 y; Kid/KiddoKINDL: parents 7–17 y. The KINDL questionnaire consists of 24 items each on a 5-point Likert scale (0 = never, 1 = rarely, 2 = sometimes, 3 = often, and 4 = always) and 6 dimensions: physical well-being, psychological well-being, self-esteem, family, friends and functioning in everyday life (school or preschool/kindergarten). To provide comparison, the total score is calculated by counting the scores given to the items of each dimension, converting them to a scale between 0–100 and adding up the results for the respective dimensions (Ünalp et al., 2022). Higher values reflect a higher health-related quality of life (Ravens-Sieberer and Bullinger, 1998a, 1998b).

The KINDL questionnaire was selected because it is the standard instrument for measuring quality of life in patients at our clinic. On the one hand, this is since there is already comparable data for recording the HRQoL of children with epilepsy based on the KINDL (Strzelczyk et al., 2019) and that this questionnaire is available free of charge. Another tool that could be used to measure HRQoL would be the ‘Quality of Life in Childhood Epilepsy Questionnaire: QOLCE-55’ (Goodwin et al., 2015), which is only a parent-proxy report, or the ‘PedsQL’ questionnaire, in which only the parent and child versions of the age group 8–12 y is freely available (Varni and Limbers, 2009).

2.5. Statistical analysis

Qualitative data collection was done using Microsoft Excel 2024® (Microsoft Corp., Redmond, USA). All statistical analyses were conducted using SPSS 29.0 software (IBM SPSS, Chicago, IL, USA). The normality of metric demographic parameters (age at testing, age at epilepsy onset, epilepsy duration) was tested using the Kolmogorov-Smirnov test. An independent samples *t*-test (when variables were normally distributed) and two Mann-Whitney *U* tests (when variables were not normally distributed) were used to analyse our two groups (‘unimpaired’ vs. ‘impaired’) for possible differences in the demographic parameters mentioned above. For testing nominal and ordinal distributed demographic data for significant differences several Chi-Square tests and Fisher’s Exact tests were utilized. As it was found that there were significant differences in some demographic data between the two groups, further statistical tests were conducted to prevent selection bias. More precisely, Spearman’s correlation was used to analyze possible correlations between demographic data (epilepsy onset, epilepsy duration, seizure frequency, number of ASMs) and the three KINDL subscores with significant differences between our groups (‘self-esteem’ and ‘family’ child reported, ‘friends’ parent reported). To assess the association between executive functioning and HRQoL independent sample *t*-tests were performed for subscores with normal distribution (‘physical well-being’ child reported, ‘total score’ child reported, ‘total score’ parent reported) and several Mann-Whitney *U* tests were used for subscores that did not fit normally. Spearman’s correlation was conducted to explore the association between EpiTrack scores and KINDL subscores and KINDL total score. The effect sizes were calculated either with Cohen’s *d* for continuous variables or with Cramér’s *V* for categorical variables, depending on the measurement level of the individual variables. In general, effect sizes were interpreted according to standard conventions: small (Cohen’s *d* $\approx$ 0.2; Cramér’s *V* $\approx$ 0.1), medium ($\approx$ 0.5 / 0.3), and large ($\geq$ 0.8 / 0.5) (Cohen, 1992; Schober et al., 2021). The Benjamini-Hochberg correction procedure for multiple testing was used for all statistical analyses. The two-tailed + significance level was accepted as $p < 0.05$.

3. Results

3.1. Patient disposition and demographic data

231 possible candidates were screened for eligibility, of which 112 remained for final analysis due to further exclusion criteria (Fig. 1). Those were separated into two groups (children with and without executive dysfunction as measured by the EpiTrack Junior).

A detailed description of the reasons not to participate in this study was summarized in the supplementary file. A summary of the socio-demographic data of the participants and their distribution was presented in Table 1. Significant differences were found between the two cohorts for age at epilepsy onset ($p = 0.001$; adjusted *p*-level: 0.005; $d = 0.61$) and epilepsy duration ($p = 0.002$; adjusted *p*-level: 0.007; $d = 0.58$) unveiling children with lower EpiTrack scores (≤ 28 P) developed epilepsy at an earlier age and therefore had a longer duration of illness. Furthermore, children with impaired executive function had a significantly higher seizure frequency (adjusted *p*-value: 0.03; $V = 0.30$) and were treated with more ASMs (adjusted *p*-value: 0.005; $V = 0.36$). Although some differences reached statistical significance, the predominantly low to moderate effect sizes (according to conventional threshold values) allow a cautious interpretation. No significant differences were found for any of the remaining demographic variables and CBCL. There was no significant correlation between CBCL scores and EpiTrack scores.

A series of Pearson’s and Spearman’s correlations were used to investigate the relationship between the age of epilepsy onset, epilepsy duration, seizure frequency or number of ASMs taken and values of the KINDL subscales ‘self-esteem’, ‘friends’ and ‘family’. No significant correlation was found between these demographic data and these dimensions. This applies to both the cohort of children with and without executive dysfunction.

3.2. Association between executive functions and perceived quality of life

In both reports (child and parents) versions, children with an impaired executive function in the EpiTrack Junior did not show a poorer perceived quality of life based on the total score of KINDL questionnaire than those with normal executive function. For the parents’ version children with EpiTrack scores on average had significant higher values for the ‘friends’ dimension than those with impaired scores (Mdn = 75.0 vs. 62.5, respectively), U ($N_1 = 64$, $N_2 = 45$) = 852.5; $z = -3.645$; adjusted *p*-value: 0.007; $r = 0.349$ (Fig. 2a). The children’s version showed that participants without executive dysfunction had higher scores for the ‘family’ dimension than those with executive dysfunctions (Mdn = 78.5 vs. 75.0, respectively), U ($N_1 = 58$, $N_2 = 38$) = 737.5; $z = -2.759$; adjusted *p*-value: 0.042; $r = 0.281$ (table 3b; Fig. 2b). In the children’s version, in the group without executive dysfunction the value of the dimension ‘self-esteem’ was statistically lower than in the executive impaired group (Mdn = 62.5 vs. 75.0, respectively), U ($N_1 = 58$, $N_2 = 38$) = 767.0; $z = -2.524$; adjusted *p*-value: 0.042; $r = 0.258$ (Fig. 2b).

There was a weak positive correlation between the total EpiTrack score and the KINDL ‘friends’ dimension, parent report ($r = 0.285$; $n = 109$; adjusted *p*-value: 0.021, Table 2). Moreover, there was a weak positive correlation between the total EpiTrack score and the KINDL ‘family’ dimension, child report ($r = 0.276$; $N = 96$; adjusted *p*-value: 0.021, Table 2) and a weak negative correlation between the total EpiTrack score and the KINDL ‘self-esteem’ dimension, child report ($r = -0.289$; $N = 96$; adjusted *p*-value: 0.021, Table 2).

4. Discussion

In this study we investigated the role of executive dysfunction on the HRQoL in children and adolescents with epilepsy.

Previous studies used mainly parent-rated questionnaires (Schraegle

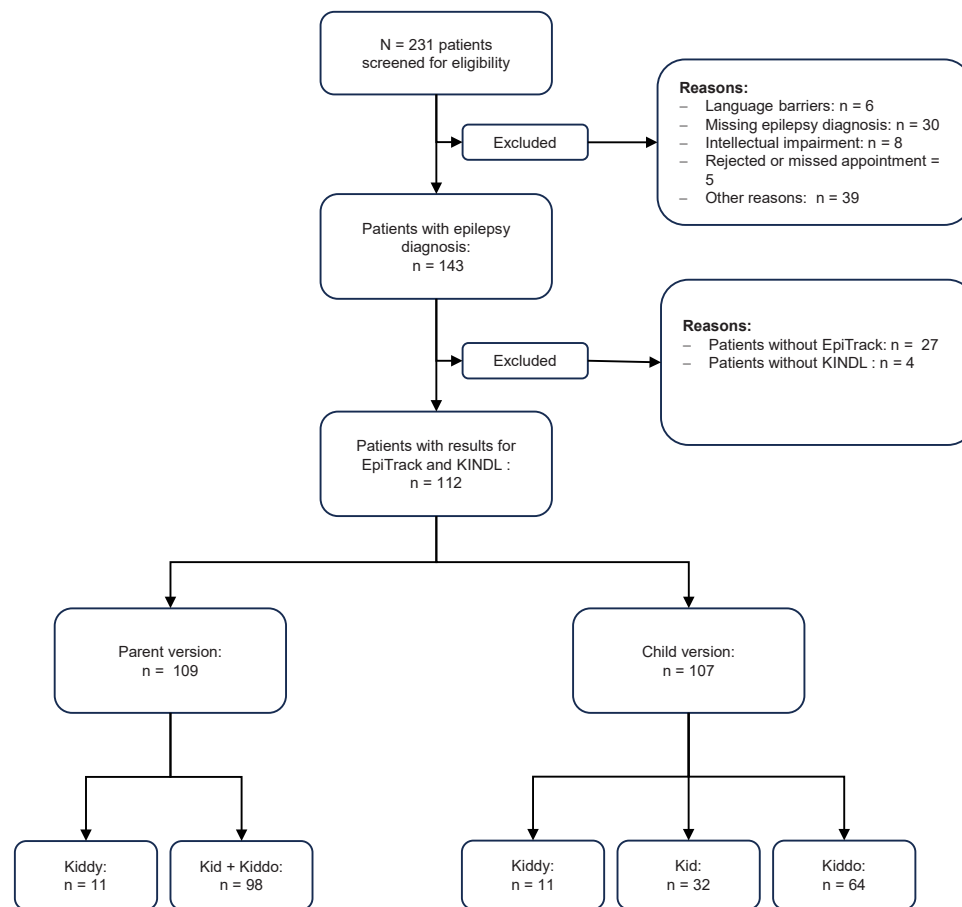


Fig. 1. Flow chart of patient disposition showing step-by-step selection process with inclusion and exclusion criteria.

and Titus, 2016; Sherman et al., 2006; Zanaboni et al., 2024). In our study we used the KINDL questionnaire, which also considers the child's perspective and abilities. Our goal was to supplement previous studies dealing with this topic and especially focusing on both perspectives, the child's perspective and the parents' perspectives.

Significant correlations between executive dysfunction and the KINDL total score, representing the entire HRQoL, as well as most sub-dimensions, were not present, which could indicate that there is no obvious relationship between EF and HRQoL.

Nevertheless, a possible association between executive dysfunction and reduced HRQoL was found in some subcategories. HRQoL was reduced in the areas of family and friends due to an increased occurrence of conflict, disharmony, and the feeling of not being 'well received' by others in children with executive dysfunction. In contrast, children with executive dysfunction had significantly higher self-esteem scores than children without dysfunction. These results suggest that executive functions are more strongly associated with certain aspects of HRQoL and could act as so-called mediating factors.

A well-established model that focuses on such mediators influencing self-assessed HRQoL is the 'Wilson and Cleary Model of Health-Related Quality of Life (HRQOL)'. It is a conceptual framework model that links the relationship between biological and physiological aspects, clinical symptoms, and functional status in order to understand the effects of disease on self-assessed HRQoL (Sousa and Kwok, 2006). Thus, in this framework, executive functions could be understood as a biological factor mediating the relationship between a chronic disease such as epilepsy and the self-assessed HRQoL in children and adolescents (van de Loo et al., 2022).

In the 'family' category, children with executive dysfunctions reported more often that there were bad arguments at home and that they

felt less comfortable in their family environment and got on less well with their parents compared to children without executive dysfunction. It is already known that the effects of epilepsy are far-reaching and extend beyond the person suffering to the whole family unit (Thompson and Upton, 1992; Ellis et al., 2000). A few studies that have focused on HRQoL in families with CWE have described that disruption of the family could be observed more often if neuropsychological disorders were also present in children with epilepsy (Ostendorf and Gedela, 2017; Thornton et al., 2008). This could explain the reduced HRQoL found in our study in families with children with executive dysfunction.

Children in our impaired group spent according to their parents less time with friends, argued more with them and were less likely to feel 'well received' by their friends. Executive functions are described, among other things, to be responsible for self-regulatory functions, as well as the ability to organize and plan situations and to regulate behavioral and emotional responses (Seniów, 2012; Sung and Choi, 2021; Gioia et al., 2001). This highlights the importance of EF for responsible social behaviour (Lezak, 1993) and could explain why the children in the impaired group in our study report a higher burden in this area.

A number of clinical studies, especially of a retrospective nature, have used questionnaires (e.g. BRIEF) to assess executive functions, which carry the risk of subjective bias. In fact, some data even show that the BRIEF and EpiTrack junior do not correlate well with each other (Hauger et al., 2023). In the current study, we used the EpiTrack Junior, a validated executive function test that was administered to the patient and thus allows a better objective assessment of executive functions than a questionnaire.

Psychiatric and behavioral problems are prevalent in populations of children and adolescents with epilepsy (Aaberg et al., 2016) and they are

Table 1

Demographic characteristics of the patients with and without impairment of executive functions.

Total	n	Executive function not impaired	Executive function impaired	p-values (ES (d/V/Φ))
EpiTrack Groups	112	67 (59.8 %)	45 (40.2 %)	
Sex (n)				
m/f	52/60	32/35	20/25	0.811 (0.04)
Average age at testing in years				
Mean (95 %CI); SD	112	11.8 (12.8–10.9); 3.8	11.5 (10.5–12.4); 3.1	0.772 (0.08)
Average age at epilepsy onset in years				
Mean (95 %CI); SD	112	7.5 (8.6–6.4); 4.5	5.0 (3.9–6.1); 3.6	0.005** (0.61)
Average epilepsy duration in years				
Mean (95 %CI); SD	112	4.1 (5.0–3.2); 3.7	6.4 (7.6–5.2); 4.1	0.007** (0.58)
Seizure frequency (n)				
Free	60	44 (73.3 %)	16 (26.7 %)	0.03* (0.30)
Min 1x/ last months	20	10 (50.0 %)	10 (50.0 %)	
1x months	5	1 (20.0 %)	4 (80.0 %)	
Weekly	15	7 (46.7 %)	8 (53.3 %)	
Daily	10	4 (40.0 %)	6 (60.0 %)	
Seizure type (n)				
Focal	60	32 (53.3 %)	28 (46.7 %)	0.386 (0.18)
Generalized	49	31 (63.3 %)	15 (36.3 %)	
Unclassified	3	2 (66.7 %)	1 (33.3 %)	
Unknown onset	3	2 (66.7 %)	1 (33.3 %)	
Epilepsy type (n)				
Focal	70	37 (52.8 %)	33 (47.2 %)	0.09 (0.26)
Generalized	35	26 (74.3 %)	9 (25.7 %)	
Combined generalized and focal	1	0 (0.0 %)	1 (100.0 %)	
Unknown	6	4 (66.7 %)	2 (33.3 %)	
Epilepsy Etiologies				
Genetic	46	28 (60.8 %)	18 (39.2 %)	0.208 (0.23)
Structural	31	14 (45.2 %)	17 (54.8 %)	
Metabolic	2	2 (100.0 %)	0 (0.0 %)	
Immunologic	1	0 (0.0 %)	1 (100.0 %)	
Infectious	1	1 (100.0 %)	0 (0.0 %)	
Unknown	31	22 (70.9 %)	9 (29.1 %)	
Epilepsy Syndroms				
Neonate/Infant	5	3 (60.0 %)	2 (40.0 %)	0.864 (0.07)
Childhood	27	16 (59.3 %)	11 (40.7 %)	
Variable Age	3	1 (33.3 %)	2 (66.7 %)	
ASM (n)				
0	17	12 (70.6 %)	5 (29.4 %)	0.005** (0.36)
1	58	42 (72.4 %)	16 (27.6 %)	
2	25	9 (36.0 %)	16 (64.0 %)	
≥ 3	12	4 (33.3 %)	8 (66.7 %)	
Behavioral status				
CBCL	101	43/17 (28.3 % of patients had behavioral abnormalities)	23/18 (43.9 % of patients had behavioral abnormalities)	0.14
behavior unaffected: CBCL < 64/behavior affected: CBCL ≥ 64				

CI: confidence interval; SD: standard deviation; ASM: anti-seizure medication
ES: effect size

*p-value is significant at the 0.05 level (2-tailed)

**p-value is significant at the 0.01 level

Significant data are marked in bold

Significance following Benjamini-Hochberg correction procedure

known to be associated with both quality of life (Schraegle and Titus, 2021) as well as executive dysfunction (Alfstad et al., 2016). Subjective reports of executive dysfunctions may correlate with emotional factors, e.g. measured by depressive symptoms or symptoms of stress (Samarasekera et al., 2015; Ten Eycke and Dewey, 2016). Although there were more patients with an abnormal behaviour score (CBCL ≥ 64) in the patient group with executive dysfunction, this difference was not statistically significant compared to the group of patients without executive dysfunction. Nevertheless, it cannot be ruled out that the increased proportion of patients with behavioural problems in the group with executive dysfunctions has a bias on the perceived quality of life and thus also the results. Beyond that, we have no further details on differences in specific neuropsychiatric problems in the two groups (EF dysfunction vs. normal EF). Nevertheless, we used the CBCL to measure behavioural problems, which may not cover the full spectrum of neuropsychiatric problems.

Notably, the children with executive dysfunction showed greater self-esteem despite lower scores in the categories ‘family’ (self-rated) and ‘friends’ (parent-rated). They stated that they liked themselves better, felt more comfortable in their skin and were more often proud of themselves and their ideas. So far, there is hardly any literature on the connection between executive functions and self-esteem. However, there are already some studies that have investigated the relationship between cognitive abilities and self-esteem and their results may be transferable to our findings. Children with higher intelligence were described as more sensitive to the judgment of others, on which their self-esteem depends (Papadopoulos, 2021). The fact that executive functions (also measured with the EpiTrack) correlate well with intelligence scores underlines the possible link between impaired intelligence and higher self-esteem in patients with epilepsy (Hauger et al., 2023; Helmstaedter et al., 2019).

In addition, self-critical thinking and negative metacognition, as well as the feeling of being different, are associated with lower self-esteem (Sherman et al., 2006; Papadopoulos, 2021). With an impairment of executive functions, children may have limited insight into the impact of their reduced executive abilities and are less aware of external evaluation (Netson et al., 2016; Hoge and Renzulli, 1993).

Another explanation for the reciprocal association of self-esteem and executive functions is that self-esteem helps the child to cope with cognitive problems and thus reduces the negative effect on quality of life. Therefore, enhancing self-esteem is an important goal of psychological interventions in patients with epilepsy (Cahill et al., 2021).

This could explain the higher level of self-esteem found in children with executive dysfunction in our study.

The group with executive dysfunction had significantly more ASMs, longer duration of epilepsy, earlier epilepsy onset and fewer were seizure free. The further analysis of these correlations is challenging, as they are also strongly interdependent. Patients with early manifestation often have a more severe course of epilepsy that is difficult to treat and requires polytherapy and are nevertheless rarely seizure-free. In addition, they have a significantly higher risk of cognitive impairment than patients with a later manifestation (Sorg et al., 2022). Furthermore, studies show negative effects of cumulative ASMs on cognition (Witt et al., 2015).

Although there were significant differences between our two cohorts in terms of seizure frequency, number of ASMs, epilepsy onset and duration, no direct influence of those demographics on well-being in terms of family, friends and self-esteem was found.

From this, we can deduce that the different scores for HRQoL found cannot be attributed, at least not directly, to the differences in the epilepsy-related data of our two cohorts and are much more likely to be related to factors such as the level of executive functions (Sherman et al., 2006; Love et al., 2016). However, the impact of some epilepsy-related parameters on the expression of executive functions should be considered. Some studies have already shown that an increase in seizure frequency, duration of epilepsy and the number of current ASMs, as well as

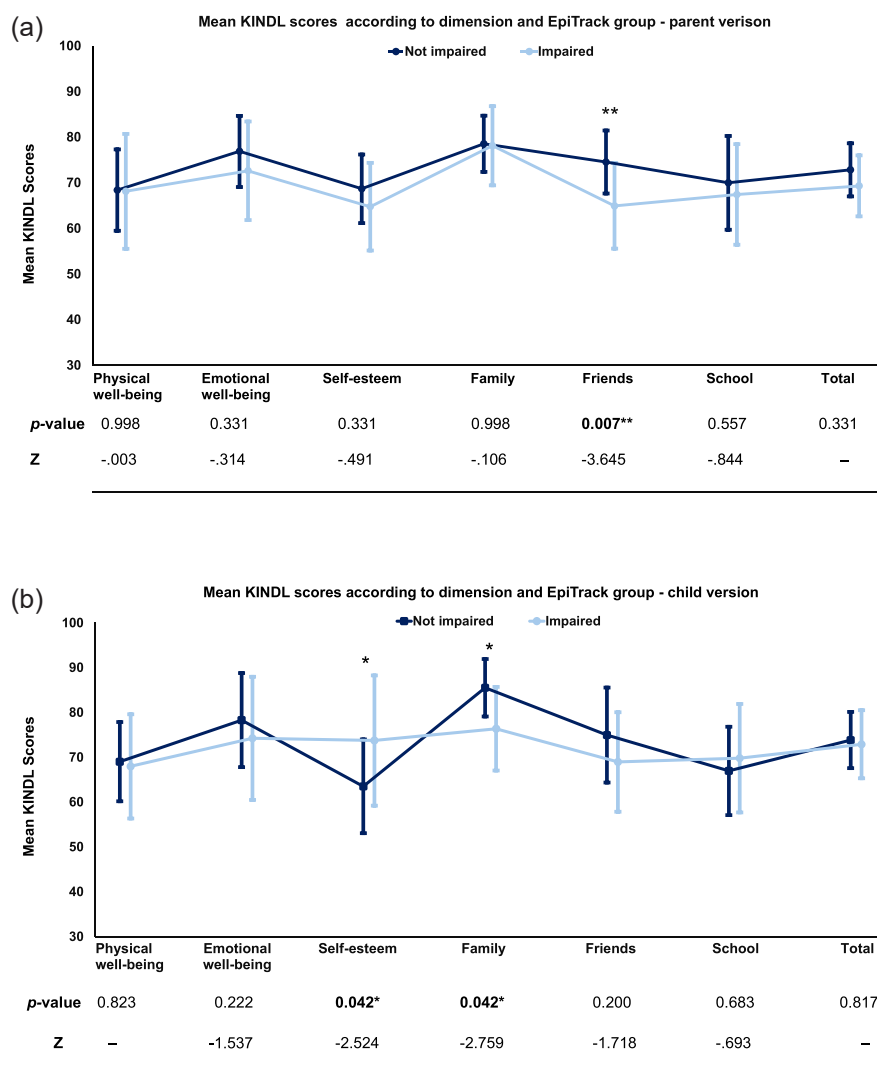


Fig. 2. a-b: Error bar chart for mean KINDL scores according to KINDL dimension and EpiTrack group. Error bars represent the 95 % confidence interval. Tables below contain values of the independent-sample *t*-test and Mann-Whitney *U* test. If no Z-value was given, an independent-sample *t*-test was performed. Independent variable: EpiTrack result (≤ 28 P and > 28 P9). a): parent version, b): child version; **p*-value is significant at the 0.05 level (2-tailed), ***p*-value is significant at the 0.01 level.

Table 2

Values of Spearman's rank correlation (r_s) between KINDL scores and EpiTrack scores; parent and child version.

parent and child version							
	Physical well being	Emotinal well being	Self esteem	Family	Friends	School	Total
EpiTrack Score							
<i>Parent version</i>	.022	.124	.145	.066	.285*	.123	.195
<i>Child version</i>	.053	.201	–.289*	.276*	.203	.016	.097

Significance following Benjamini-Hochberg procedure

Signifikant data are marked in bold

*. Correlation is significant at the 0.05 level (2-tailed).

a young age at onset is associated with a decline in executive functions (Black et al., 2010; Thompson and Duncan, 2005; Cainelli et al., 2021). It can therefore be assumed that, as indicated by the results of our study, depending on the severity of these epilepsy-related parameters, a more severe epilepsy, as well as more executive dysfunction and, consequently, a reduced HRQoL, can be expected. It should also be noted that although no direct correlation was found between these epilepsy-related data and HRQoL in the areas of family, friends and self-esteem, it cannot

be concluded that these factors may directly influence the general HRQoL of children with epilepsy. This was shown by are other studies that suggest that the higher the number of ASMs taken (Johnson et al., 2004) or the longer the duration of epilepsy (Edefonti et al., 2011), the lower the overall quality of life. However, the influence on HRQoL of specific areas of life was not described.

5. Limitations

When interpreting the results of our study, a few limitations should be taken into consideration. The first limitation is the selection of participants from a tertiary epilepsy center rather than the general population. There could therefore be a selection bias, as patients who are cared for in such facilities may have a more complex clinical picture with a higher seizure burden and more comorbidities than those who are cared for in outpatient practices. Another factor limiting our study was the heterogeneous group size of our two cohorts, with a slightly larger number of participants in the group of children without executive dysfunctions. It must also be considered that not all participants were able to perform all neuropsychological tests. This was partly since they were too young or, as it turned out in the course of the study, they were too cognitively impaired or did not have the necessary knowledge of German, which is why in these cases the KINDL questionnaire was only completed by the parents. The KINDL questionnaire was also sometimes only filled out by the children, for example, because the parents did not have time or did not speak German well enough. Finally, it should be mentioned that the 4–6 y version of the KINDL questionnaire only provides an overall score for the HRQoL and no additional scores for the individual dimensions. The EpiTrack Junior is primarily a short screening test for clinical use. Therefore, the transfer of data to other developmental and cognitive domains must be interpreted with caution. Nevertheless, the EpiTrack has shown that it correlates well with overall IQ, so that it allows a more than indicative assessment of cognitive performance in patients with epilepsy (Helmstaedter et al., 2019). The data presented are based on a survey conducted as part of a cross-sectional study. Statements about the intra-individual course can therefore only be made in the context of a longitudinal survey, which should be part of future investigations. Another limitation of the work is the lack of systematic, detailed neuropsychological testing in the patient cohort. Although, as mentioned above, IQ scores correlate well with executive function (including EpiTrack), this screening test is not a substitute for a detailed neuropsychological evaluation. Although the data show a relationship between executive functions and quality of life in some dimensions, they must be interpreted with caution because confounders such as a detailed neuropsychological evaluation and family factors were not recorded.

6. Conclusion

No obvious association was found between executive functions and the KINDL total score and thus the overall perceived HRQoL within the investigated cohort. However, executive dysfunction appeared to be associated with a higher risk of lower HRQoL in the areas of family and friends. Self-esteem, on the other hand, indicated to be increased in children and adolescents with an impairment of executive functions. Social interactions, family environment and self-esteem thus seemed to be the most vulnerable HRQoL domains depending on the state of executive functions.

CRedit authorship contribution statement

Leonie Grosse: Writing – original draft. **Malin Zaddach:** Writing – original draft. **Christine Makowski:** Writing – review & editing, Methodology. **Moritz Tacke:** Formal analysis. **Miriam Gerstner:** Methodology, Investigation, Data curation, Conceptualization. **Ingo Borggräfe:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Robert Optiz:** Formal analysis, Data curation. **Mirjam N Landgraf:** Writing – review & editing. **Elisabeth Kaufmann:** Writing – review & editing, Supervision. **Muehe Lena:** Writing – original draft, Methodology, Investigation, Formal analysis.

Acknowledgement

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Supplementary file

Detailed description of reasons not to participate in this study.

Conflict of interest

“The authors have no financial or non-financial interests to disclose. We confirm that we have read the journal’s position on issues involved in ethical publication and affirm that this report is consistent with those guidelines. The study protocol (#21–0887) was confirmed by the local ethical board of the Medical Faculty of the University of Munich.”

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.epilepsyres.2025.107646](https://doi.org/10.1016/j.epilepsyres.2025.107646).

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